Data Analytics report

Jul 29, 2026, 1:07 PM UTC

Curve CryptoSwap vs current DH2, with CPMM control

Matched invariant-shape, common-controller, and native-controller economic comparison of Curve CryptoSwap and current DH2, with CPMM retained as the zero-concentration control.

Executive summary

At the equal-execution scalar cross-section, **Current DH2** has the numerically lowest one-jump LP loss. At the equal-jump-risk scalar cross-section, **Current DH2** has the numerically lowest execution index. These are boundary-conditioned results from the declared sampled domains—not Pareto results or global family optima. Equal-execution LP-loss scalar leader status: **Current DH2** is **boundary-censored** within `peak_depth = 2…2,000; fade_scale = 1e−6…0.5`. Equal-jump-risk execution scalar leader status: **Current DH2** is **boundary-censored** within `peak_depth = 2…2,000; fade_scale = 1e−6…0.5`. The common-controller paired medians favor current DH2 over the CryptoSwap invariant in both synthetic and historical classes.

Source: Corrected deterministic economic benchmark
File: results/summary.json

Native-controller result

Under the LP-net fee-adjusted ledger, neither system clears every native acceptance rule at the primary equal-execution lock across routes, flows, the top-ranked policy shortlist, pool scales, volume levels, and one-minute cadence. For the same profiles selected on fee-adjusted synthetic return, the gross development-history outcome is diagnostic only: Fixed routing favors Curve by 0.12 bps per 12 hours; Elastic routing favors Curve by 0.08 bps per 12 hours. It is not a gross universal-winner claim. Fixed routing's median fee-adjusted result favors Dusk by 0.06 bps per 12 hours. Elastic routing's median fee-adjusted result favors Dusk by 0.03 bps per 12 hours. The locked-profile LP-return-only result at the primary equal-execution invariant match favors **Current DH2 + Dusk controller** by **0.05 bps per 12 hours** at the all-cell median. No system clears the stricter 1-bp, same-direction materiality threshold in both routes. Fixed and elastic route medians are +0.06 bps and +0.03 bps; positive values favor Dusk. This ranks LP return only. Execution cost, captured flow, tracking, and the stricter universal gate remain separate. **Dusk synthetic product-readiness gate: fail.** The selected DH2 profile requires its own `rho = 2.069`; the common Curve–DH2 comparison relaxation cannot make it ready. **Sealed confirmation: PASS.** The locked development classification was **neither system**; the post-lock sealed core classification is **neither system**. They agree under the predeclared route, flow-route, and CPMM core thresholds. The original route-level training envelope was infeasible for the concentrated pair. The smallest common normalized widening was 206.9% (`rho = 2.069`). Original-envelope pass: Curve **no**, Dusk **no**. Because `rho > 1`, bounded fraction guardrails cease to bind and cost ceilings are widened by more than 2×; this shared rho governs only the fair Curve-versus-DH2 profile lock, not Dusk product readiness. The independently selected CPMM also fails its original route-service envelope. Its separate normalized widening is 170.5% (`rho = 1.705`). It remains an economic comparator, but it is not eligible for a ship or product-ready recommendation. The result is a fixed-liquidity simulation under the declared routing, flow, accounting, and robustness gates—not a live-volume forecast.

Source: Canonical fixed-liquidity native decision summary
File: results/native_summary.json

Three comparison layers keep curve shape separate from controller economics

Invariant only: At the equal-execution scalar cross-section, **Current DH2** has the numerically lowest one-jump LP loss. At the equal-jump-risk scalar cross-section, **Current DH2** has the numerically lowest execution index. These are boundary-conditioned results from the declared sampled domains—not Pareto results or global family optima. Equal-execution LP-loss scalar leader status: **Current DH2** is **boundary-censored** within `peak_depth = 2…2,000; fade_scale = 1e−6…0.5`. Equal-jump-risk execution scalar leader status: **Current DH2** is **boundary-censored** within `peak_depth = 2…2,000; fade_scale = 1e−6…0.5`.

Controlled system: with the same Dusk fees, internal EMA, order tape, retention, and center controller, current DH2 leads the CryptoSwap invariant by $2 at the synthetic paired-cell median. Current DH2 leads the CryptoSwap invariant by $167 at the historical paired-cell median. These paired medians use the common controller; they are path-dependent and do not establish a Pareto or global invariant winner.

Native system: each system keeps its own fee, EMA, profit budget, and recenter controller. Under the LP-net fee-adjusted ledger, neither system clears every native acceptance rule at the primary equal-execution lock across routes, flows, the top-ranked policy shortlist, pool scales, volume levels, and one-minute cadence.

Holding service constant makes the scalar comparison like-for-like

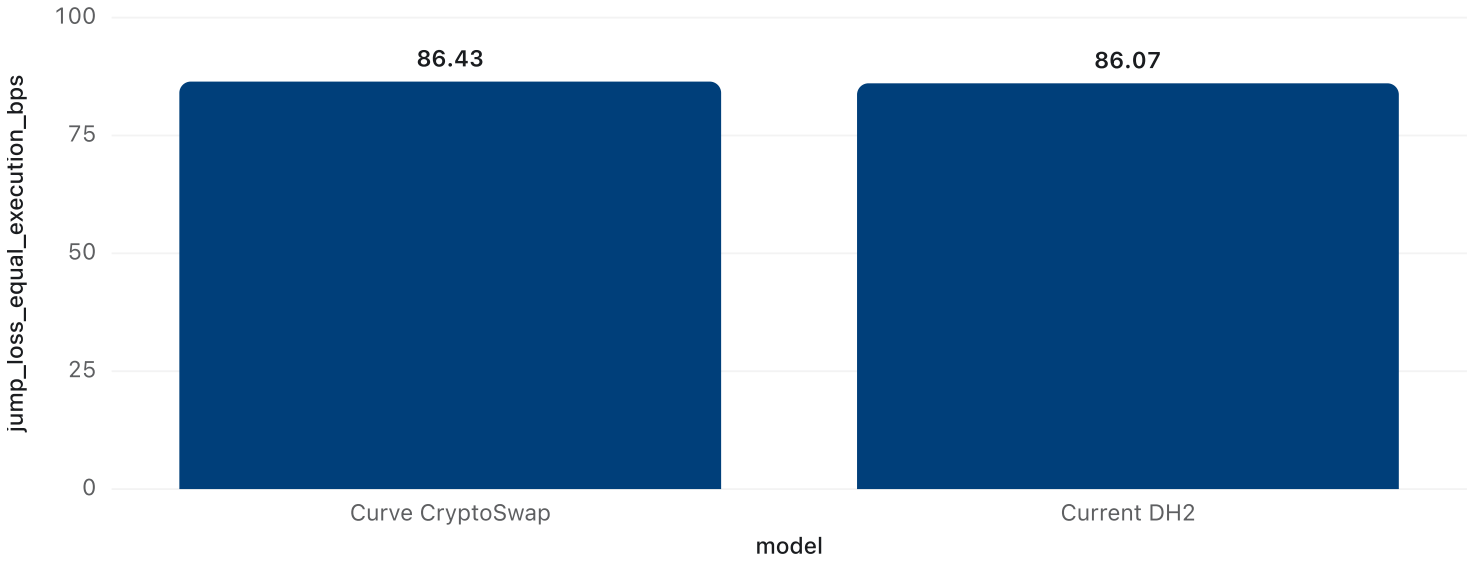
The prior ranking rewarded designs that simply quoted worse or accepted less flow. These bars hold the broad execution index equal before comparing LP jump loss. At the equal-execution scalar cross-section, **Current DH2** has the numerically lowest one-jump LP loss. At the equal-jump-risk scalar cross-section, **Current DH2** has the numerically lowest execution index. These are boundary-conditioned results from the declared sampled domains—not Pareto results or global family optima. Equal-execution LP-loss scalar leader status: **Current DH2** is **boundary-censored** within `peak_depth = 2…2,000; fade_scale = 1e−6…0.5`. Equal-jump-risk execution scalar leader status: **Current DH2** is **boundary-censored** within `peak_depth = 2…2,000; fade_scale = 1e−6…0.5`.

Declared domains: **Curve CryptoSwap:** paper A = 0.1…1,000; gamma = 1e-8…0.199; **Current DH2:** peak_depth = 2…2,000; fade_scale = 1e-6…0.5. The boundary-cell censor marks these selected scalar points: equal execution / Curve CryptoSwap, equal execution / Current DH2, equal jump risk / Curve CryptoSwap, equal jump risk / Current DH2. Their precise gaps and apparent direction are conditional on the declared domains; they are not global family optima.

Source: Corrected deterministic economic benchmark
File: results/summary.json

One-jump LP loss at equal execution service

Source: Corrected deterministic economic benchmark
Table: reference_comparison



Reference-region comparison

Source: Corrected deterministic economic benchmark
Table: reference_comparison

Curve family	Jump loss @ equal execution	Execution @ equal risk	Center impact	Equal-execution parameters	Equal-execution boundary-censored	Equal-risk parameters	Equal-risk boundary-censored	Declared search domain
Current DH2	86.07	63.77	1.95	peak_depth=2000;fade_scale=2.1348254e-05	true	peak_depth=2000;fade_scale=2.9455007e-05	true	peak_depth = 2…2,000; fade_scale = 1e-6…0.5
Curve CryptoSwap	86.43	64.7	0.35	A=1000;gamma=0.000461233335	true	A=1000;gamma=0.000620371164	true	paper A = 0.1…1,000; gamma = 1e-8…0.199

Concrete center quotes expose how the same broad service is distributed

These are zero-fee exact-input quotes for a balanced $2.0M marked-TVL pool. Each direction's cost and output value are computed independently before the two-direction mean. Matching the aggregate execution index does not force identical center slippage: the invariants distribute liquidity differently across the curve.

At an illustrative **$150/SOL** center, a 100-bps-of-TVL trade means selling either 133.333 SOL or 20,000 USDC:

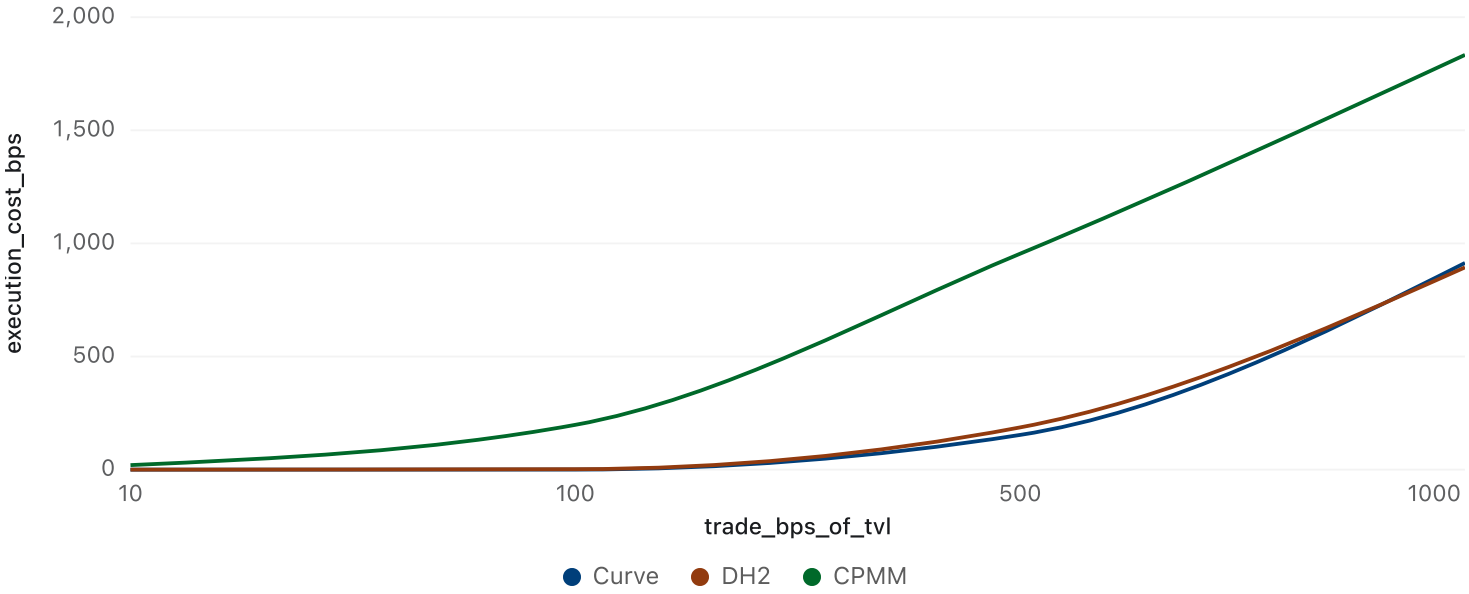
- **Curve CryptoSwap:** sell 133.333 SOL for 19,999.30 USDC (0.348 bps), or sell 20,000 USDC for 133.328699 SOL (0.348 bps).
- **Current DH2:** sell 133.333 SOL for 19,996.09 USDC (1.954 bps), or sell 20,000 USDC for 133.307274 SOL (1.955 bps).

- **CPMM:** sell 133.333 SOL for 19,607.84 USDC (196.078 bps), or sell 20,000 USDC for 130.718954 SOL (200.000 bps).

Source: Center quote and slippage sweep
File: results/shape.csv

Center exact-input slippage by trade size

Source: Center quote and slippage sweep
Table: center_quotes



Concrete center quote examples

Source: Center quote and slippage sweep
Table: center_quotes

Invariant	Size (bps TVL)	Input value (USD)	Execution cost (bps)	Quoted output value (USD)
Curve CryptoSwap	10	$2,000	0.010 bps	$2,000.00
Current DH2	10	$2,000	0.012 bps	$2,000.00
CPMM	10	$2,000	19.980 bps	$1,996.01
Curve CryptoSwap	100	$20,000	0.348 bps	$19,999.30
Current DH2	100	$20,000	1.955 bps	$19,996.09
CPMM	100	$20,000	198.039 bps	$19,607.84
Curve CryptoSwap	500	$100,000	153.479 bps	$98,476.99
Current DH2	500	$100,000	186.732 bps	$98,150.11
CPMM	500	$100,000	954.545 bps	$90,909.09
Curve CryptoSwap	1,000	$200,000	913.594 bps	$182,561.04
Current DH2	1,000	$200,000	894.077 bps	$182,916.24
CPMM	1,000	$200,000	1833.333 bps	$166,666.67

The recomputed Curve–DH2 Pareto frontier contains both ranked families

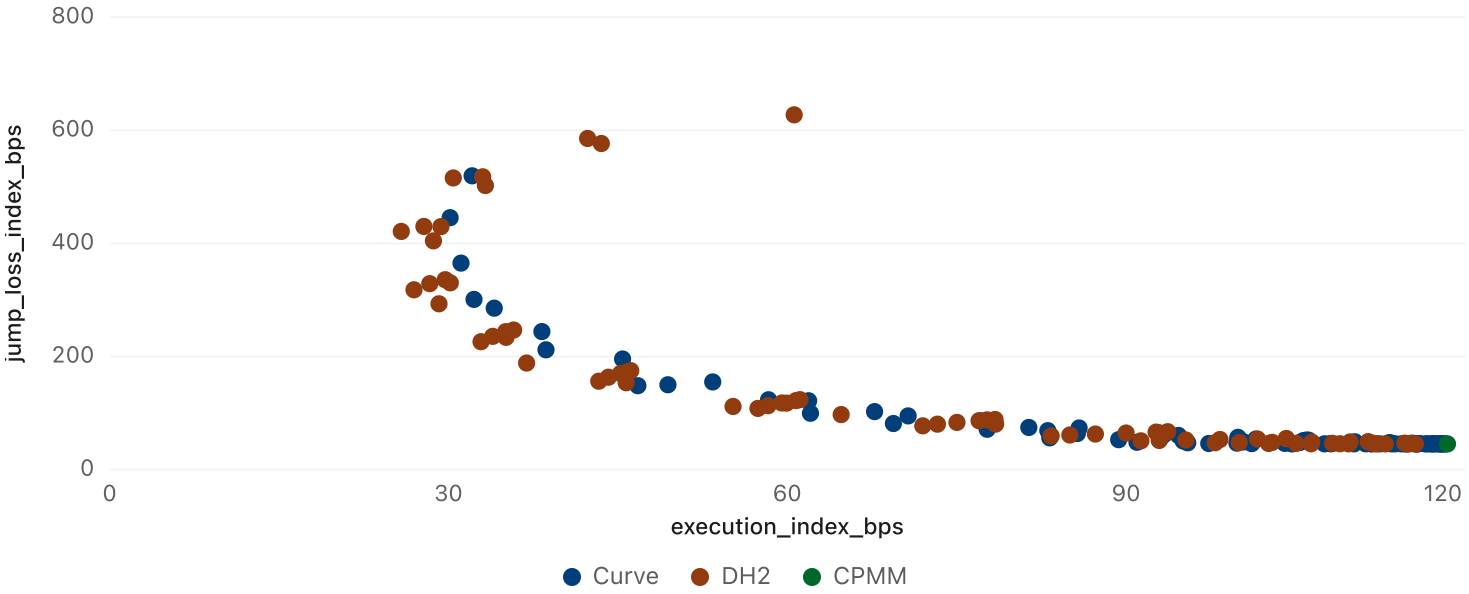
Curve CryptoSwap has 23 sampled frontier points, Current DH2 has 14 sampled frontier points. The zero-concentration CPMM control has 1 sampled frontier point. Frontier membership means only that a family has at least one non-dominated sampled configuration; point count is not a quality score. Scalar cross-section

leadership does not prove Pareto or global dominance. The lower-left envelope is boundary- and grid-conditioned.

Source: Invariant parameter sweep
File: results/pareto.csv

Execution cost versus one-jump LP loss

Source: Invariant parameter sweep
Table: pareto



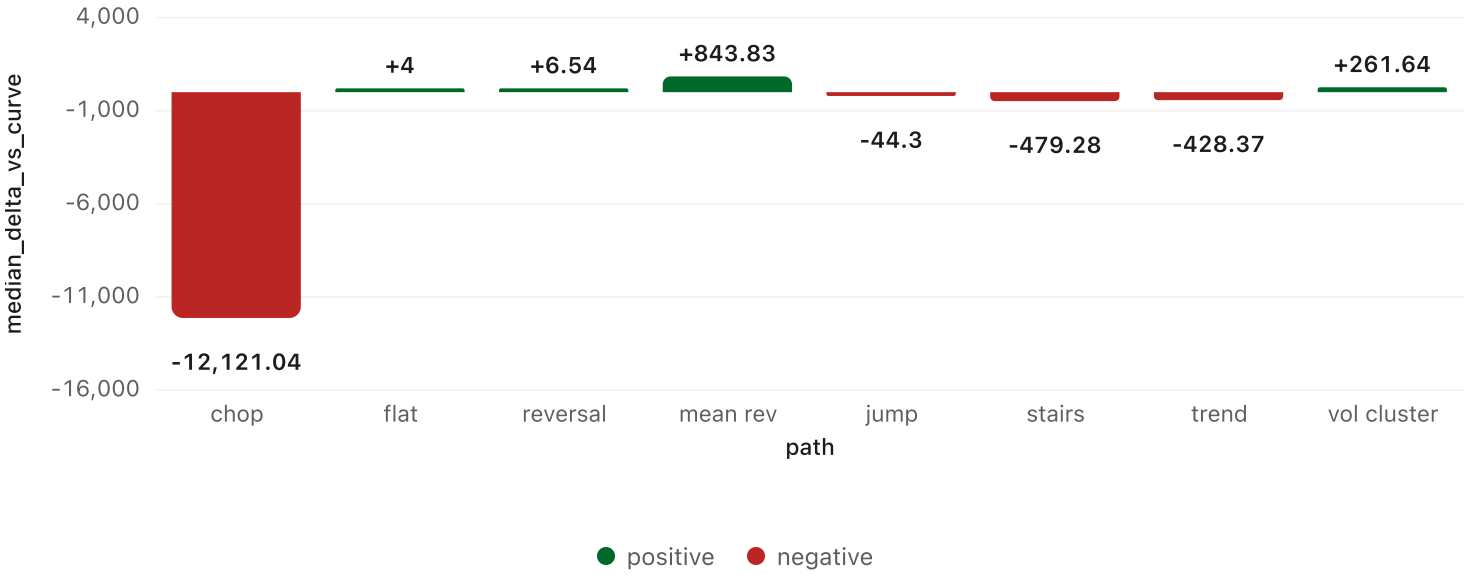
Paired controlled paths are derived pair by pair

Current DH2 leads the CryptoSwap invariant by $2 at the synthetic paired-cell median. Current DH2 leads the CryptoSwap invariant by $167 at the historical paired-cell median. These paired medians use the common controller; they are path-dependent and do not establish a Pareto or global invariant winner.

Source: Paired controlled path replay
File: results/paired_differences.csv

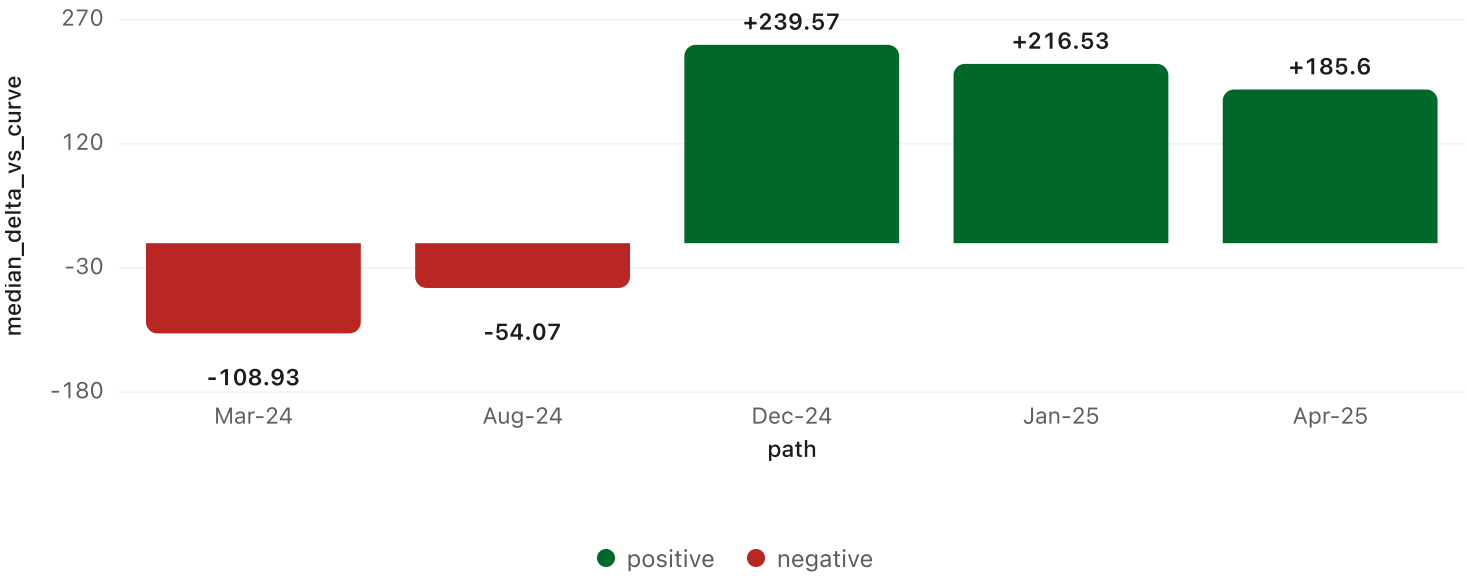
Synthetic-path median LP return difference versus Curve

Source: Paired controlled path replay
Table: path_delta_synthetic



Historical-window median LP return difference versus Curve

Source: Paired controlled path replay
Table: path_delta_historical



Quote specialization is not the same as economic dominance

The static tournament shows which curve is cheapest at each declared state and order size. A high share means a design specializes in more grid cells; it does not forecast Jupiter volume because real intent sizes and state

frequencies are unknown.

Source: Corrected deterministic economic benchmark
File: results/summary.json

Binary Curve–DH2 equal-intent quote tournament

Source: Static equal-intent quote tournament
Table: routing_summary

Curve family	Best-quote share	Median cost (bps)	Intent rows
Current DH2	50.4%	110.63	264
Curve CryptoSwap	49.6%	112.26	264

Fees and recentering now reconcile

Claimable fees remain token-denominated LP wealth but are not executable. Retained surcharge is counted once inside reserves. Divergence is outward-only, volatility comes from AMM trade shocks, and protected liquidity uses the center-normalized Q. Recenter impairment is a capacity diagnostic, not a second P&L subtraction.

Source: Corrected deterministic economic benchmark
File: results/summary.json

Synthetic profile lock does not read development history

The original route-level training envelope was infeasible for the concentrated pair. The smallest common normalized widening was 206.9% (`rho = 2.069`). Original-envelope pass: Curve **no**, Dusk **no**. Because `rho > 1`, bounded fraction guardrails cease to bind and cost ceilings are widened by more than 2×; this shared rho governs only the fair Curve-versus-DH2 profile lock, not Dusk product readiness. The independently selected CPMM also fails its original route-service envelope. Its separate normalized widening is 170.5% (`rho = 1.705`). It remains an economic comparator, but it is not eligible for a ship or product-ready recommendation.

The selected Dusk profile independently fails its original product-readiness envelope. Its own minimum normalized widening is 206.9% (`rho = 2.069`). Curve's comparative rho neither blocks nor rescues this Dusk-only gate, so the profile is not product-ready on synthetic training.

Hard numerical, finite-metric, accounting, and no-oracle gates never relax. Service uses the smallest identical normalized widening that admits Curve and concentrated Dusk; CPMM uses a separate widening and cannot contaminate that lock. Dusk product readiness uses the selected Dusk profile's own rho and is independent of both comparison rho values. Within each envelope, selection first minimizes the profile's own required rho, then maximizes worse-route fee-adjusted return utility among ties. Historical windows are not read.

Original per-route guardrails include median capture ≥50%, median selected-order cost ≤25 bps, p90 selected-order cost ≤100 bps, median marginal-tracking stale time ≤75%, and worst-flow service tails. The selected rows show each original-envelope result:

- **Curve:** original pass **no**; elastic capture 62.3%; p90 cost 280.71 bps fixed / 281.51 bps elastic; marginal-tracking stale 6.6% fixed / 1.4% elastic; 1%-TVL depth degradation (diagnostic only) 87.5% fixed / 87.5% elastic.

- **Current DH2:** original pass **no**; elastic capture 58.7%; p90 cost 306.86 bps fixed / 281.83 bps elastic; marginal-tracking stale 6.2% fixed / 0.7% elastic; 1%-TVL depth degradation (diagnostic only) 77.1% fixed / 77.1% elastic.
- **Dusk CPMM:** original pass **no**; elastic capture 51.4%; p90 cost 270.47 bps fixed / 269.52 bps elastic; marginal-tracking stale 9.7% fixed / 2.1% elastic; 1%-TVL depth degradation (diagnostic only) 9.0% fixed / 1.7% elastic.

Selected-order costs are finite-size service measures. The 1%-TVL depth-degradation probe remains a raw diagnostic; it is not the stale-price gate.

Source: Canonical fixed-liquidity native decision summary
File: results/native_summary.json

Three service-relaxation policies

Source: Canonical native selection policies
Table: native_service_policy

Policy	Required rho	rho = 0 pass	Selected profile	Authority
CPMM fallback qualification	170.5%	false	grid_c08_f10	Qualifies CPMM only; cannot change Curve–DH2 or Dusk readiness
Curve–DH2 comparative lock	206.9%	false	paired family lock	Fair profile comparison only; cannot block or rescue Dusk
Dusk product readiness	206.9%	false	grid_c09_f10	Authoritative synthetic readiness gate for Dusk

Symmetric concentrated searches and an independent CPMM search bind every promoted policy

Curve and concentrated Dusk each receive 150 initial profiles across the same 128 synthetic cells, for 19,200 decision cells per family. CPMM independently receives the same 150-by-128 budget. The audit rejects quick or partial output, stale source hashes, a missing promoted search winner, and any canonical profile outside its source-bound top-ranked policy shortlist. Search winner and canonical lock remain separate because the service envelope may select another source-ranked promoted policy.

Source: Fail-closed final report evidence receipt
File: results/final_report_audit.json

Symmetric concentrated searches, independent CPMM search, and canonical lock

Source: Final report evidence audit
Table: search_lock

Family	Full search cells	Search winner	Promoted	Canonical lock	Locked configuration
Curve Twocrypto-NG v2.1	19,200	curve_grid_c01_f02	5	curve_grid_c01_f02	A=1000; gamma=0.000461233; mid/out=5.0/5.0 bps; EMA=10.0167m; min step=1.5 bps
Current DH2 + Dusk controller	19,200	grid_c09_f10	5	grid_c09_f10	peak=2000; fade=2.13483e-05; EMA=60m; threshold/step=200.0/5

Family	Full search cells	Search winner	Promoted	Canonical lock	Locked configuration
					0.0 bps; dynamic cap=0.0 bps
Independently selected Dusk CPMM	19,200	grid_c08_f10	5	grid_c08_f10	peak=0; fade=0; EMA=30m; threshold/step=100.0/25.0 bps; dynamic cap=0.0 bps

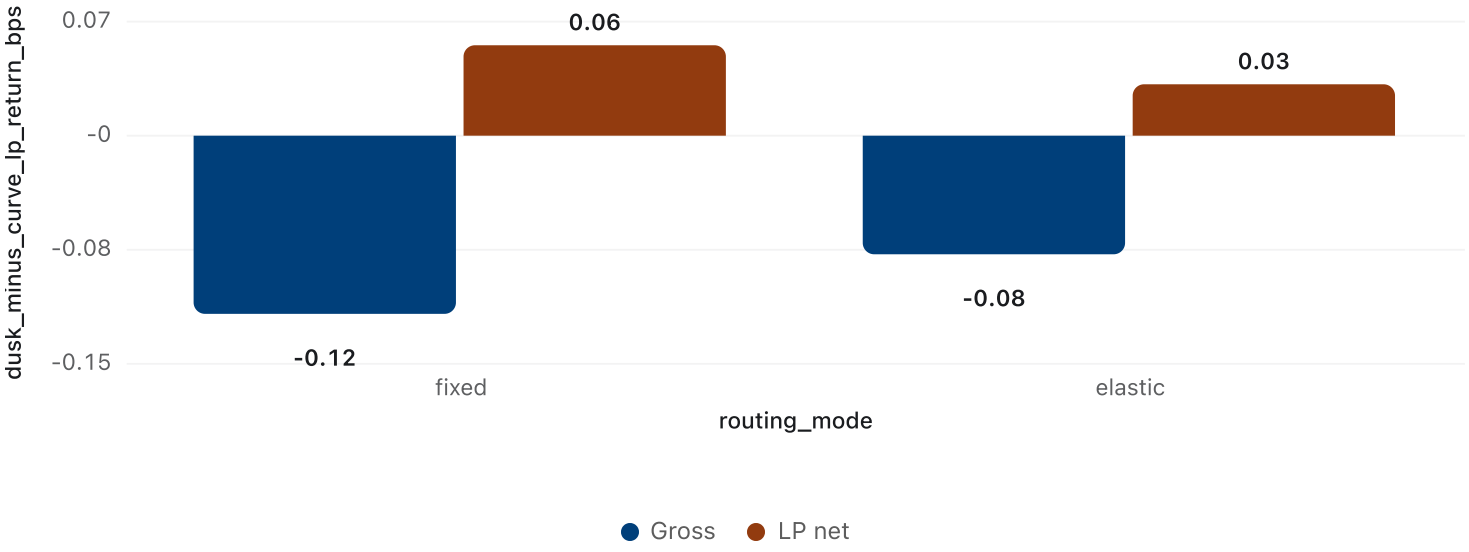
Fee-adjusted ranking keeps gross outcome diagnostic

Under the LP-net fee-adjusted ledger, neither system clears every native acceptance rule at the primary equal-execution lock across routes, flows, the top-ranked policy shortlist, pool scales, volume levels, and one-minute cadence. For the same profiles selected on fee-adjusted synthetic return, the gross development-history outcome is diagnostic only: Fixed routing favors Curve by 0.12 bps per 12 hours; Elastic routing favors Curve by 0.08 bps per 12 hours. It is not a gross universal-winner claim. Positive bars favor Dusk. Gross return is a diagnostic for the fee-adjusted-selected profiles and is measured before external revenue liabilities; fee-adjusted return applies the declared Curve admin and Dusk manager/protocol ownership while preserving liabilities already present at the score boundary.

Source: Canonical fixed-liquidity native decision summary
File: results/native_summary.json

Native Dusk-minus-Curve LP return by routing mode

Source: Canonical fixed-liquidity native decision summary
Table: native_route_return



Native route-level paired outcomes

Source: Canonical fixed-liquidity native decision summary
Table: native_route

Routing	Gross return delta (bps)	Fee-adjusted delta (bps)	p10 fee-adjusted delta	Execution-cost delta	Capture delta	Marginal-tracking stale delta	1%-TVL depth degradation delta (diagnostic)
elastic	-0.08	+0.03	-3.42	-1.62	+2.5%	0%	0%
fixed	-0.12	+0.06	-4.91	-2.89	0%	-0.7%	0%

LP-return leadership and universal dominance are separate questions

The locked-profile LP-return-only result at the primary equal-execution invariant match favors **Current DH2 + Dusk controller** by **0.05 bps per 12 hours** at the all-cell median. No system clears the stricter 1-bp, same-direction materiality threshold in both routes. Fixed and elastic route medians are +0.06 bps and +0.03 bps; positive values favor Dusk. This ranks LP return only. Execution cost, captured flow, tracking, and the stricter universal gate remain separate. The p10/p90 distribution and adjacent service deltas remain visible in the table, so a null universal winner cannot hide the measured LP-return direction.

Source: Canonical fixed-liquidity native decision summary
File: results/native_summary.json

Locked-profile LP-return leader at primary equal execution

Source: Canonical locked-profile LP-return result
Table: native_lp_leader

LP-return leader	Material both-routes leader	Median return delta (bps)	p10 return delta	p90 return delta	Dusk cell wins	Fixed-route median	Elastic-route median	Execution-cost delta	Capture delta	Marginal-stale delta
Current DH2 + Dusk controller	neither system	+0.05	-4.58	+0.49	57.5%	+0.06	+0.03	-2.22	0%	0%

Sealed post-lock history confirms or rejects replication without retuning

Sealed confirmation: PASS. The locked development classification was **neither system**; the post-lock sealed core classification is **neither system**. They agree under the predeclared route, flow-route, and CPMM core thresholds. Development history was selection-blind but visible during benchmark development; sealed history was fetched only after profiles, searches, matrix, and replay engine were locked. The confirmation uses the existing route, flow-route, and CPMM core thresholds. It cannot change the selected profile.

Source: Fail-closed final report evidence receipt
File: results/final_report_audit.json

Development-history versus sealed-history outcomes

Source: Post-lock sealed evidence audit
Table: sealed_route_comparison

Route	Development return (bps)	Sealed return (bps)	Development window wins	Sealed window wins
fixed	0.06	0.09	66.7%	60%
elastic	0.03	0.05	60%	60%

Audit correction preserved without changing economics

During the evidence audit, the first sealed-summary producer was found to take the maximum signed accounting residual instead of the maximum absolute residual. The economic result and paired ledgers were unchanged; their preserved hashes are `7017fccfb0c059eaa230dafa916f852fda528ff5ce8b22b01f8317b0b8900109` and `408b1078298b1a6a7c1c38a575556bf4009ccb2c6fafff5ffd4a19218171cb8`. The corrected producer

is covered by a mixed-sign regression, and the true maximum absolute residual is 2.081e-07, below the 1.0e-05 limit.

Source: Sealed accounting-summary correction receipt
File: results/sealed_accounting_summary_correction.json

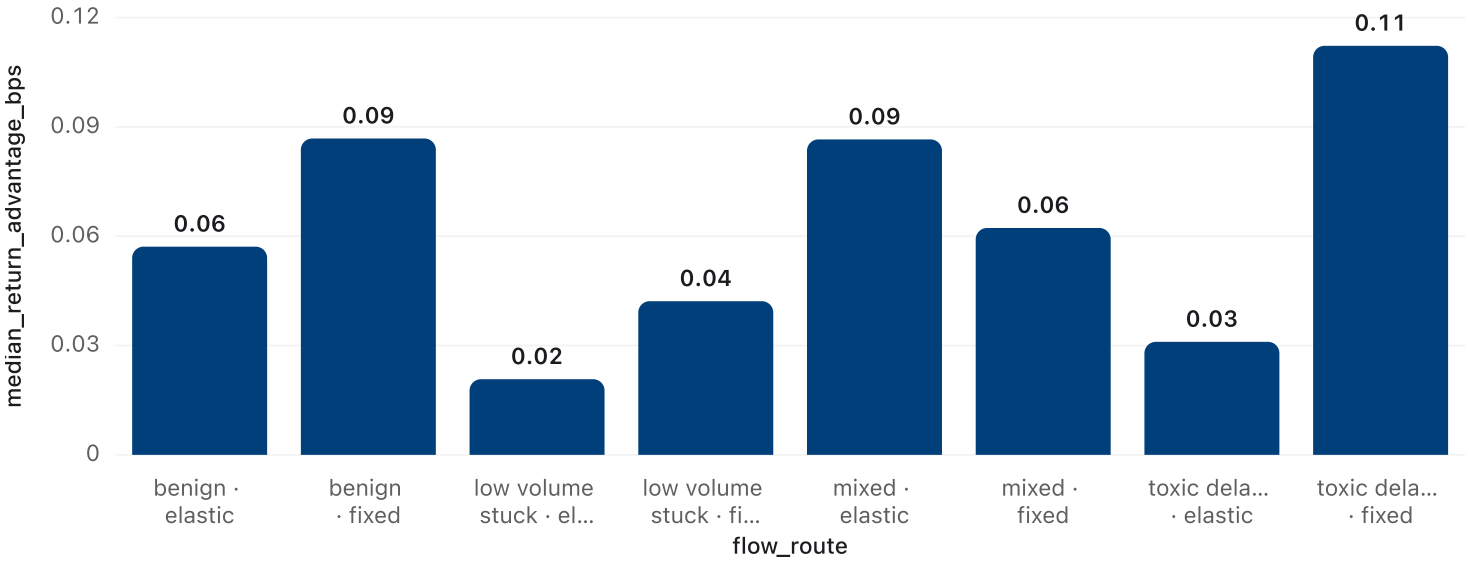
Route and flow tails—not a pooled median—control the native decision

Each flow-by-route cell must avoid a median return reversal, and its return, selected-order execution, capture, and marginal-tracking stale tails must stay inside the declared limits. This chart shows the median fee-adjusted return delta; the decision also uses the richer tail fields preserved in the report dataset.

Source: Canonical fixed-liquidity native decision summary
File: results/native_summary.json

Native fee-adjusted LP return by flow and route

Source: Canonical fixed-liquidity native decision summary
Table: native_flow_route



Alternate invariant matches are locked-controller diagnostics

The native replay is repeated under nearest full quote-profile, equal one-jump risk, and pinned v2.1 deployment-invariant matching while retaining the controllers and fees selected on equal execution. These are non-optimized transfer diagnostics; a reversal here cannot grant, revoke, or reverse the primary universal label.

Source: Canonical fixed-liquidity native decision summary
File: results/native_summary.json

Locked-controller alternate invariant-match diagnostics

Source: Native locked-controller matching-basis diagnostics

Table: native_basis

Basis	Route	Median return delta	Bottom-decile mean	p90 execution regression	p10 capture advantage	p90 marginal-stale regression
equal_jump_risk	elastic	+0.01	-11.01	-0.09	-1.4%	2.2%
equal_jump_risk	fixed	+0.05	-11.89	0.95	0%	0.7%
nearest_full_profile	elastic	-0.04	-18.52	-0.59	-0%	2.8%
nearest_full_profile	fixed	-0.05	-18.84	-1.71	0%	0.7%
v21_deployed_invariant	elastic	+0.16	-1.23	16.07	-28.9%	0.7%
v21_deployed_invariant	fixed	+0.56	-0.7	28.68	0%	8.4%

Locked profiles are tested from 5% to 400% daily volume

Notional throughput changes while arrival frequency, profiles, fees, and controllers stay locked. This tests transfer rather than reoptimizing a different winner for each volume. The universal gate reads every underlying flow-route cell; the table reports compact medians and worst tails.

Source: Canonical fixed-liquidity native decision summary

File: results/native_summary.json

Locked-profile transfer across daily-volume regimes

Source: Native locked-profile daily-volume sensitivity

Table: native_volume

Comparison	Daily volume / pool	Median of cell medians	Worst bottom-decile mean	Worst p90 execution regression	Worst p10 capture advantage	Worst p90 marginal-stale regression
Curve – CPMM	5%	-0.06	-39.6	0.85	-16.6%	3.5%
Dusk – CPMM	5%	+0.01	-40.1	-0.94	-0%	3.8%
Dusk – Curve	5%	+0.07	-0.62	0.76	-0.4%	2.1%
Curve – CPMM	25%	-0.11	-40.02	25.04	-13.6%	10.8%
Dusk – CPMM	25%	+0.02	-40.46	7.51	-10.4%	9.4%
Dusk – Curve	25%	-0.06	-0.78	0.97	-3.5%	4.2%
Curve – CPMM	100%	+0.1	-60.48	92.2	-26.3%	7.3%
Dusk – CPMM	100%	-0.05	-42.28	91.68	-25.6%	10.8%
Dusk – Curve	100%	+0.47	-3.18	37.09	-4.6%	17%
Curve – CPMM	400%	-10.37	-76.33	79.69	-10.1%	-1.4%
Dusk – CPMM	400%	+3.97	-67.46	92.39	0%	20.8%
Dusk – Curve	400%	+14.9	-70.78	121.03	-9.8%	39.2%

Revenue ownership can change LP ranking without changing quotes

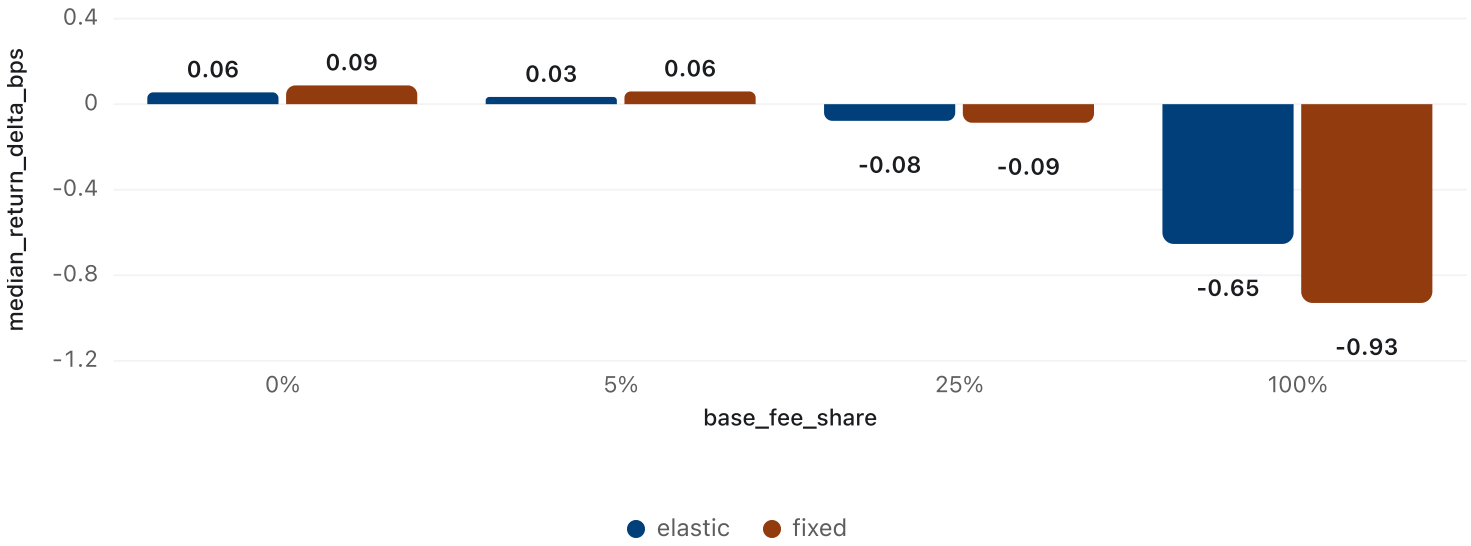
The primary ledger assigns 5% of Dusk base fees to the manager, keeps all Dusk dynamic surcharge with LPs, and applies Curve's 25% effective gross-xcp-profit liability. The sensitivity revalues Dusk ownership without altering any trade, quote, reserve, or controller path.

Source: Canonical fixed-liquidity native decision summary

File: results/native_summary.json

Native LP return under alternate Dusk base-fee ownership

Source: Dusk base-fee ownership sensitivity
Table: native_fee_split



Center-movement counterfactual isolates the value and cost of recentering

Enabling center movement changes median LP return by +0.00 bps for Curve and +0.00 bps for Dusk across paired fixed/elastic route, flow, seed, and window cells. Execution, capture, marginal tracking, and tracking changes are reported alongside return so recentering is not credited for P&L while hiding a service regression.

Source: Native center-movement-off counterfactual
File: results/native_controller_counterfactual.csv

Native center movement on minus off effects

Source: Native center-movement-off counterfactual
Table: native_controller

System	LP return delta	Execution-cost delta	Capture delta	Marginal-tracking stale delta	Tracking-RMSE delta
Current DH2 + Dusk controller	0	0	0%	0%	0
Curve Twocrypto-NG v2.1	+0	-0.13	0%	0%	-14.47

Curve can reopen same-slot arbitrage after recentering

A deterministic regression witness required **20 profitable fills and 19 price-scale recenters** before no profitable follow-up remained. It converged without a numerical failure. The benchmark therefore uses a Curve-local emergency ceiling of 64 fills; the shared DH2/CPMM ceiling remains 16. This is a controller-mechanics witness, not a claimed canonical-matrix maximum or an on-chain execution count.

Source: Deterministic Curve same-slot cascade regression
File: tests/test_curve_native.py

CPMM comparator and invariant ablation answer different questions

The product CPMM independently selects profile `grid_c08_f10` on synthetic training. Against it, current DH2 changes median LP return by -0.01 bps and selected execution cost by -8.48 bps. The independently selected CPMM also fails its original route-service envelope. Its separate normalized widening is 170.5% (`rho = 1.705`). It remains an economic comparator, but it is not eligible for a ship or product-ready recommendation. This comparison gates the universal claim. The matched-policy row preserves current DH2's selected fees/controller and zeros only `peak_depth` and `fade_scale`; it isolates concentration and remains diagnostic.

Source: Canonical fixed-liquidity native decision summary

File: results/native_summary.json

CPMM product comparator versus concentration ablation

Source: Canonical native CPMM comparison summaries

Table: native_cpmm

Comparison	CPMM profile	CPMM service rho	Product status	Winner gate	LP return delta	Cell-win share	Execution-cost delta	Capture delta	Marginal-tracking stale delta
Current DH2 minus independently selected CPMM	grid_c08_f10	170.5%	not ready	true	-0.01	47.9%	-8.48	0%	-0.7%
Current DH2 minus matched-policy CPMM	grid_c09_f10	—	not applicable	false	-0.01	47.9%	-8.48	0%	-0.7%

Universal means every primary equal-execution predicate passes

The table preserves the complete fee-adjusted pass/fail audit for both systems. Gross outcomes do not receive a universal label. A failed predicate blocks the fee-adjusted universal label even when the headline median is positive. `No universal winner` therefore means insufficient dominance under the declared gate, not equal economic performance. Alternate invariant matches are reported separately and do not enter this gate.

Source: Canonical fixed-liquidity native decision summary

File: results/native_summary.json

Native universal-winner acceptance predicates

Source: Canonical fixed-liquidity native decision summary

Table: native_predicates

Economics view	Candidate	Required predicate	Pass
Fee-adjusted LP universal gate	Current DH2 + Dusk controller	all arbitrage cascades converged before hard cap	true
Fee-adjusted LP universal gate	Current DH2 + Dusk controller	all arbitrage searches numerically valid	true
Fee-adjusted LP universal gate	Current DH2 + Dusk controller	selected profile passes original training service envelope	false
Fee-adjusted LP universal gate	Current DH2 + Dusk controller	both concentrated profiles pass original training service envelope	false
Fee-adjusted LP universal gate	Current DH2 + Dusk controller	training service is deployment ready	false
Fee-adjusted LP universal gate	Current DH2 + Dusk controller	common training service relaxation is zero	false
Fee-adjusted LP universal gate	Current DH2 + Dusk controller	both routes median advantage at least 1bp	false
Fee-adjusted LP universal gate	Current DH2 + Dusk controller	both routes window win fraction at least 70pct	false
Fee-adjusted LP universal gate	Current DH2 + Dusk controller	both routes bottom decile mean not below minus 1bp	false
Fee-adjusted LP universal gate	Current DH2 + Dusk controller	both routes tail service regressions within limits	false

Economics view	Candidate	Required predicate	Pass
Fee-adjusted LP universal gate	Current DH2 + Dusk controller	every flow route median has no return reversal	true
Fee-adjusted LP universal gate	Current DH2 + Dusk controller	every flow route bottom decile mean not below minus 1bp	false
Fee-adjusted LP universal gate	Current DH2 + Dusk controller	every flow route tail service regressions within limits	false
Fee-adjusted LP universal gate	Current DH2 + Dusk controller	top ranked policy shortlist win fraction at least 60pct	false
Fee-adjusted LP universal gate	Current DH2 + Dusk controller	every top ranked policy shortlist flow route has no reversal	false

56 results · Showing first 15

Decision-grade references pass; the sharp fade is hard rejected

Curve, current DH2, and CPMM pass the common numerical sweep. The fixed exponent-2 sharp-DH2 experiment fails a stronger uniqueness check: one reserve state has three invariant roots. Current DH2 is cross-validated against Rust; the rejected sharp experiment is analytical only.

Source: Corrected deterministic economic benchmark
File: results/summary.json

Common reference checks

Source: Common numerical reference checks
Table: safety

Model	Solver failures	Non-monotonic quotes	Max invariant error	Invariant roots	Hard rejected	Arithmetic scope
CPMM	0	0	0	1	false	high_precision_analytical_reference
Current DH2	0	0	0	1	false	rust_fixed_point_fixture_validated_with_10ppm_quote_haircut
Curve CryptoSwap	0	0	0	1	false	high_precision_analytical_reference
Hyperbolic p4 DH2 (experiment)	0	0	0	1	false	analytical_experiment_not_rust_validated
Sharp DH2 (rejected)	11	4	0.36	3	true	analytical_experiment_not_rust_validated

Fixed sharper fade: do not implement

The experiment keeps only `peak_depth` and `fade_scale`; exponent 2 is protocol-fixed. It decays faster only after its shoulder, but the stronger D-dependence creates three valid invariant roots in the committed witness. That is a hard safety rejection before LP-return comparisons.

Source: Sharp-DH2 multiple-root safety witness
File: results/sharp_root_evidence.csv

D-independent p4 repair: economically rejected

The D-independent p4 repair has one invariant root in every scanned state, but this remains a sampled analytical-float check rather than a fixed-point proof.

- Jump loss at equal execution: current DH2 **86.07 bps**; p4 **104.08 bps** — 20.9% higher, favoring current DH2.
- Execution index at equal jump risk: current DH2 **63.77 bps**; p4 **76.71 bps** — 20.3% higher, favoring current DH2.

Both like-for-like scalar objectives favor current DH2, so the p4 repair is economically rejected in this experiment. The evidence-derived production status is `keep_current_dh2_fade`.

Source: D-independent p4 iso-service comparison
File: results/hyperbolic_p4_comparison.csv

Decision: controlled paths favor current dh2; keep native and scalar claims separate

The common-controller paired medians favor current DH2 over the CryptoSwap invariant in both synthetic and historical classes. At the equal-execution scalar cross-section, **Current DH2** has the numerically lowest one-jump LP loss. At the equal-jump-risk scalar cross-section, **Current DH2** has the numerically lowest execution index. These are boundary-conditioned results from the declared sampled domains—not Pareto results or global family optima. Equal-execution LP-loss scalar leader status: **Current DH2** is **boundary-censored** within `peak_depth = 2…2,000; fade_scale = 1e−6…0.5`. Equal-jump-risk execution scalar leader status: **Current DH2** is **boundary-censored** within `peak_depth = 2…2,000; fade_scale = 1e−6…0.5`. For the product-level replay, under the LP-net fee-adjusted ledger, neither system clears every native acceptance rule at the primary equal-execution lock across routes, flows, the top-ranked policy shortlist, pool scales, volume levels, and one-minute cadence. **Dusk synthetic product-readiness gate: fail.** The selected DH2 profile requires its own `rho = 2.069`; the common Curve–DH2 comparison relaxation cannot make it ready. Retain exact CPMM as the zero-concentration comparator and keep the common comparison rho separate from Dusk's own product-readiness decision.

Native conclusions remain simulation-conditioned

The native result uses 15 development historical price windows plus 15 post-lock sealed windows, but user arrivals, information, outside routing, and arbitrage remain modeled. The scalar cross-sections use these declared domains: **Curve CryptoSwap:** paper A = 0.1…1,000; gamma = 1e-8…0.199; **Current DH2:** peak_depth = 2…2,000; fade_scale = 1e-6…0.5. The boundary-cell censor marks these selected scalar points: equal execution / Curve CryptoSwap, equal execution / Current DH2, equal jump risk / Curve CryptoSwap, equal jump risk / Current DH2. Their precise gaps and apparent direction are conditional on the declared domains; they are not global family optima. The independent Curve reference covers fixed-parameter swap, fee, EMA, profit gate, and price-scale updates; it does not simulate liquidity-event hooks, parameter ramps, or the liquidity-operation-triggered claim that is time-gated to at most once per 24 hours; no automatic daily transaction is assumed. Signed routed orders and markouts are still required for a Jupiter-volume claim.